

MingYu Lu

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Github: <https://github.com/q8888620002>

Google scholar: <https://scholar.google.com/citations?hl=en&pli=1&user=Hp7cv5kAAAAJ>

Education

University of Washington

Seattle, WA

Ph.D. in Computer Science, Advisor: Su-In Lee

Sept 2021 - Winter 2027 (expected)

- Courses include Machine Learning, Interactive Learning, Randomized Algorithm, and Computational Biology.

Harvard Medical School

Boston, MA

M.S. in Biomedical Informatics.

May 2019

- Thesis *Sensitivity Analysis of Deep Q-Learning for Sepsis Treatment*.

- Courses include Computational Statistics, Biostatistics, Machine Learning, and Data Science in Medicine.

Kaohsiung Medical University

Kaohsiung, Taiwan

Doctor of Medicine.

June 2017

- Studies included biochemistry, genetics, pharmacology, pathology, anatomy, physiology and microbiology.

Honors

LEAP* Fellowship of the Ministry of Science and Technology of Taiwan 2019

Research

AIMS Lab, University of Washington

Seattle, WA

Experience

Research Assistant

Sept 2021 - Present

- Developing advanced machine learning methods at the intersection of explainable AI, feature and data attribution, generative models, and treatment effect estimation. These methods are applied within the biomedical domain to improve understanding and support decision-making in complex real-world healthcare settings.

Bosch Research

Sunnyvale, CA

Research Intern

June 2025 - Sep 2025

- Developing cross-layer attentive probes to identify concepts in vision-language models (VLMs), and benchmarking their understanding for autonomous driving.

MIT Laboratory for Computational Physiology

Cambridge, MA

Visiting Scholar

June 2023 - Sep 2023

- Develop contrastive learning methods for time series prediction for in-hospital pediatric infection.

Philips Research North America

Cambridge, MA

Clinical Data Scientist

2020 - 2021

- Develop scalable and efficient algorithms for clinical decision support and treatment outcome prediction. Analyzed electronic medical record and hospital workflow.

MIT Laboratory for Computational Physiology

Cambridge, MA

Postdoctoral/Graduate Research

2018 - 2020

- Developed machine learning model and evaluation metrics on dynamical treatment regimes, causal inference, counterfactual simulation models, and reinforcement learning in clinical decision making.

Academia Sinica, Institution of Information Science

Taipei, Taiwan

Research Assistant

2017 - 2018

- Improved efficiency of protein spectrum viewer by refactoring data structure and deploying visitor pattern of fragmented spectrum. Designed and implemented user interface of the spectrum viewer.

Work

Clinical Data Scientist Philips Research North America (2020 - 2021)

Experience

Postdoc Researcher MIT (2019- 2020)

Medical Intern National Taiwan University Hospital (2016 - 2017)

Publications

Mingyu Lu, Ethan Weinberger, Su-In Lee “CellCLIP - Learning Perturbation Effects in Cell Painting via Text-Guided Contrastive Learning” *NeurIPS* 2025

Mahtab Bigverdi, Wisdom Oluchi Ikezogwo, Kevin Minghan Zhang, Hyewon Jeong, **MingYu Lu**, Sungjae Cho, Linda Shapiro, Ranjay Krishna “MedBLINK: Probing Visual Perception and Trustworthiness in Multimodal Language Models for Medicine” (*Oral*) *Computer Vision for Automated Medical Diagnosis (CVAMD) at ICCV* 2025

Yubin Kim, Zhiyuan Hu, Hyewon Jeong, Eugene W Park, Shuyue Stella Li, Chanwoo Park, Shiyun Xiong, **MingYu Lu**, Hyeonhoon Lee, Xin Liu, Daniel McDuff, Cynthia Breazeal, Samir Tulebaev, Hae Won Park “BehaviorSFT: Behavioral Token Conditioning for Clinical Agents Across the Proactivity Spectrum” *EMNLP findings* 2025

Yubin Kim, Hyewon Jeong, Chanwoo Park, **MingYu Lu**, Eugene W Park, Haipeng Zhang, Xin Liu, Hyeonhoon Lee, Daniel McDuff, Cynthia Breazeal, Samir Tulebaev, Hae Won Park “Tiered Agentic Oversight: A Hierarchical Multi-Agent System for AI Safety in Healthcare” *Multi-Agent Systems (MAS) in the Era of Foundation Models at ICML* 2025

Mingyu Lu, Chris Lin, Chanwoo, Kim, Su-In Lee “An Efficient Framework for Crediting Data Contributors of Diffusion Models” *ICLR* 2025

Yubin Kim, Hyewon Jeong, Shan, Chen, Shuyue Stella Li, **Mingyu Lu** et al “Medical Hallucination in Foundation Models and Their Impact on Healthcare” *MedRxiv* 2025

• LEAP is a governmental talent program that matches top-notch Taiwanese professionals to research opportunities in the leading research institute in U.S., France and Israel. Exclusively for applicants with an M.D. degree or Ph.D. degree and significant academic achievement.

Mingyu Lu, Chris Lin, Su-In Lee “Efficient Shapley Values for Attributing Global Properties of Diffusion Models to Data Groups” *Data Problem in Foundation Models at ICLR 2024*

Mingyu Lu, Ian Covert, Nathan White, Su-In Lee “CODE-XAI: Construcing and Deciphering Treatment Effect via Explainable AI using Real-world Data” *medRxiv Under review Nature Health 2024*

Ian Covert, Wei Qiu, **Mingyu Lu**, Nayoon Kim, Nathan White, Su-In Lee “Greedy Information Maximization for Online Feature Selection” *ICML 2023*

Elias Baedorf Kassis, Stephanie Hu, **MingYu Lu**, Alistair Johnson, Somnath Bose, Maximilian S Schaefer, Daniel Talmor, H Lehman Li-wei, Zach Shahn “Titration of Ventilator Settings to Target Driving Pressure and Mechanical Power” *Respiratory Care 2023*

Yale Chang, Corneliu Antonescu, Shreyas Ravindranath, Junzi Dong, **Mingyu Lu**, Francesco Vicario, Lisa Wondrely, Pam Thompson, Dennis Swearingen, and Deepak Acharya “Early Prediction of Cardiogenic Shock Using Machine Learning” *Front. Cardiovasc. Med. - Heart Failure and Transplantation 2022*

Mingyu Lu, Yifang Chen, Su-In Lee. “A Deep Bayesian Bandits Approach for Anticancer Drug Screening : Exploration via Functional Prior” *Adaptive Experimental Design and Active Learning in the Real World at ICML 2022*

Rui Li, Stephaine Hu, **MingYu Lu**, ,Yuria Utsumi, Prithwish Chakraborty, Daby M. Sow, Piyush Madan, Jun Li, Mohamed Ghalwash, Zach Shahn, and Li-wei H. Lehman. “G-Net: a Recurrent Network Approach to G-Computation for Counterfactual Prediction Under a Dynamic Treatment Regime” *Machine Learning for Health (ML4H) at NeurIPS 2021*

MingYu Lu, Zach Shah, Finale Doshi Velez, and Li-Wei H Lehman: “Is Deep Reinforcement Learning Ready for Practical Applications in Healthcare? A Sensitivity Analysis of Duel-DDQN for Hemodynamic Management in Sepsis Patients Reinforcement Learning for Sepsis Treatment. “ *AMIA 2020. (Distinguished Paper, 5 out of 1,350)*

Niklas Rindtorff, **MingYu Lu**, Nisarg Patel, Huahua Zheng, and Alexander D’Amour: “A Biologically Plausible Benchmark for Contextual Bandit Algorithms in Precision Oncology Using in vitro Data” *Machine Learning for Health (ML4H) Workshop at NeurIPS 2019.*

Teaching

The University of Washington

Teaching Assistant CSE 543 Machine Learning 2023, 2024

Teaching Assistant CSE 527 Computational Biology 2022

Harvard-MIT

Teaching Assistant HST953 Collaborative Data Science in Medicine 2019

Service

Organizer

NewInML at NeurIPS 2020

Program committee

ICLR Navigating and Addressing Data Problems for Foundation Models (DPFM) Workshop 2025

Reviewer

NeurIPS 2025

ICLR Learning Meaningful Representation for Life (LMRL) Workshop 2025

ICLR Navigating and Addressing Data Problems for Foundation Models (DPFM) Workshop 2025

NeurIPS Datasets & Benchmark 2024

ACM Conference on Health, Inference, and Learning 2020

AMIA 2020

NeurIPS Machine Learning for Health (ML4H) Workshop 2019, 2020

Mentorship

Society of Critical Care Medicine Datathon, Milan Critical Care Datathon 2020

Beth Israel Deaconess Medical Center Artificial Intelligence/Machine Learning Datathon 2020

University and Department Service

Chief Information Officer, KMU Class of 2017

President of Guitar Club Leader, Kaohsiung Medical University (2013 - 2014)

CoFounder

TinyNote <https://thetinynotes.com/> (2016 - 2019)

A website of physician-authored clinical decision support resources, allowing medical professionals to follow the more than 1500+ latest guidelines of diseases and clinical inquiry with monthly 180,000+ active users.

- Responsible for AWS deployment, development and maintenance of back-end APIs, database, text-searching package of NodeBB, and Google search engine optimization.

Skills

Programming/Scripting Languages: Python, R, JavaScript, Java, php, C#, HTML, CSS.

Data analysis/Machine learning: Numpy, Scikit-Learn, Pandas, Tensorflow, Pytorch.

Database/Query: Postgre, MySQL, MongoDB, BigQuery.

Cloud/Web Services/Framework: AWS, GCP, Nginx, NodeJS, Express.

Virtual Environment: Docker, OpenAI Gym/Universe, Anaconda.

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